29 A neutron collides with a $^{235}_{92}U$ nucleus. The fission reaction that occurs splits the nucleus into $^{141}_{56}Ba$, $^{92}_{36}Kr$ and three neutrons, and energy is released.

The equation for this reaction is shown.

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3{}^{1}_{0}n$$

The table shows the masses of the particles in the reaction.

nucleus	mass / u
$^{235}_{92}U$	235.04
$^{141}_{56}Ba$	140.91
$^{92}_{36}Kr$	91.91
$^{1}_{0}n$	1.01

What is the energy released in this fission reaction?

A 3.0×10^{-11} J

B 1.8×10^{-10} J

C 1.8×10^{16} J

D 1.1×10^{17} J

30 Thorium-228 has a decay constant of $1.15 \times 10^{-8}\,s^{-1}$.

A small sample of thorium-228 is prepared, containing 5.00×10^{12} nuclei.

How long does it take for 3.00×10^{12} of these nuclei to decay?

A 3.62×10^{7} s **B** 4.44×10^{7} s **C** 7.23×10^{7} s **D** 7.97×10^{7} s

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CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level
Higher 2

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PHYSICS

9749/02

Paper 2 Structured Questions

October/November 2020

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

The use of an approved scientific calculator is expected, where appropriate.
Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

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[Turn over

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Answer all questions in the spaces provided.

1 (a) A cyclist travels at constant speed along horizontal ground. Frictional forces oppose the motion of the cyclist.

Use Newton's first law of motion to explain why the cyclist travels at constant speed.

.....

.....

.....

..... [2]

(b) The drag force F acting on the cyclist at speed v is given by

$$F = \frac{1}{2} c_D \rho A v^2$$

where c_D is the drag coefficient, ρ is the density of air and A is the frontal area of the cyclist.

Data for the moving cyclist is shown in Table 1.1.

Table 1.1

quantity	magnitude	uncertainty
F/N	22	±2
c_D	0.88	±0.01
ρ/kg m^{-3}	1.2	±0.1
A/m^2	0.32	±0.02

Determine the speed of the cyclist, with its actual uncertainty. Give your answer to an appropriate number of significant figures.

speed = ... ± ..... m s^{-1} [4]

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(c) (i) Derive, from the definition of work done, an expression for power output for the cyclist in terms of velocity v.

[2]

(ii) Calculate the power output P of the cyclist at the speed calculated in **(b)**.

P = ... W [1]

[Total: 9]

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2 **(a)** Define *gravitational potential* at a point.

.....

..... [1]

(b) The Earth, of radius 6400 km, has mass 6.0×10^{24} kg.

A global positioning satellite of mass 1600 kg is in an orbit of radius 2.7×10^{7} m about the centre of the Earth. The satellite is launched from the Earth's surface.

Calculate the change in gravitational potential energy ΔE_P for this satellite.

ΔE_P = ... J [2]

(c) **(i)** A satellite of mass m orbits a planet of mass M. The mass of the planet may be assumed to be a point mass at its centre.

Show that the kinetic energy E_k of the satellite in an orbit of radius r is given by

$$E_k = \frac{GMm}{2r}.$$

[2]

(ii) Use the information in **(b)** and the expression in **(c)(i)** to calculate the kinetic energy E_K of the global positioning satellite in its orbit.

E_K = ... J [1]

(iii) A student wants to calculate the change in kinetic energy of a satellite launched from the Earth to its orbit. The student calculates the initial and final kinetic energies using the expression in **(c)(i)**.

Suggest and explain whether the student's method is correct.

.....

.....

..... [2]

[Total: 8]

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3 Two parallel plates are in a vacuum. One plate is positively charged and the other plate is earthed.

A rectangular conductor of width x is placed midway between the plates so that its faces are a distance x from each plate, as shown in Fig. 3.1.

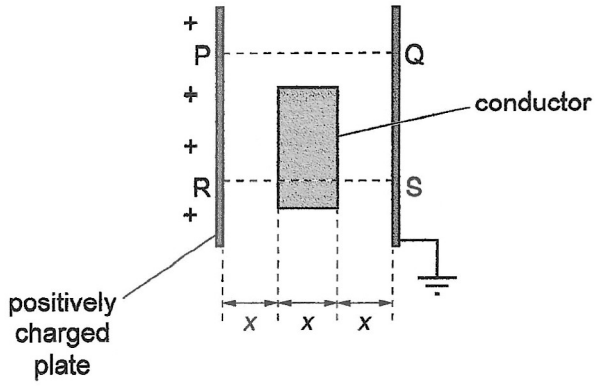


Fig. 3.1

(a) The variation with distance from P to Q of the electric potential along line PQ is shown in Fig. 3.2.

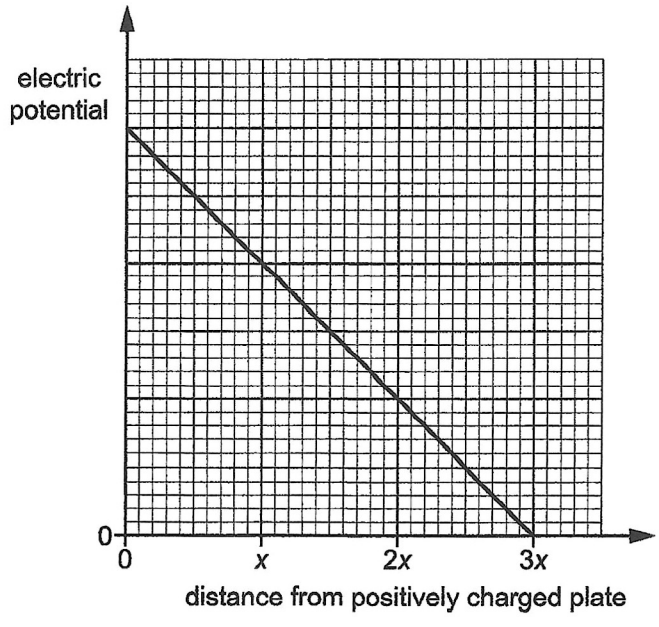


Fig. 3.2

On Fig. 3.2, draw a line to show the variation with distance from R to S of the electric potential along the line RS. [2]

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(b) The variation with distance from P to Q of the electric field strength along line PQ is shown in Fig. 3.3.

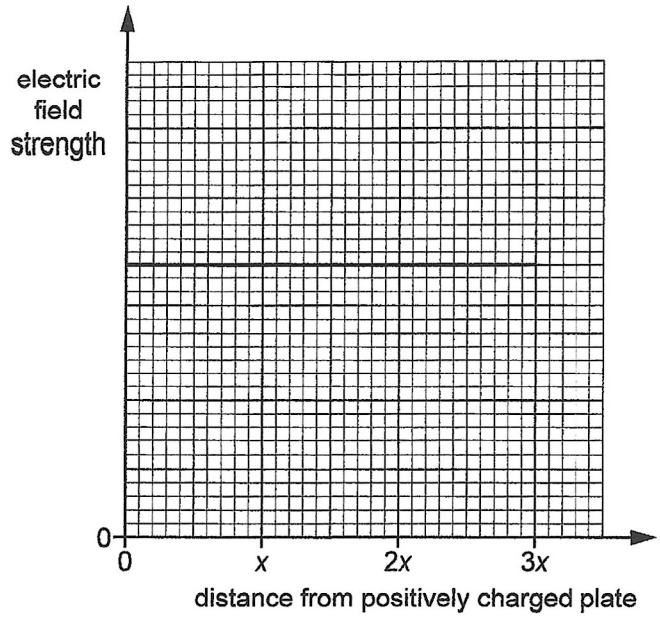


Fig. 3.3

On Fig. 3.3, draw a line to show the variation with distance from R to S of the electric field strength along the line RS. [3]

[Total: 5]

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4 **(a)** Explain what is meant by the term *diffraction*.

.....

..... [1]

(b) State two conditions necessary for two-source interference fringes to be observed.

1. .....

2. .....

[2]

(c) A laser is used to produce two-source interference fringes on a screen.

The incident laser light is normal to the plane of a double slit. The plane of the double slit is parallel to the screen.

The variation of the fringe width x with distance D between the double slit and the screen is shown in Fig. 4.1.

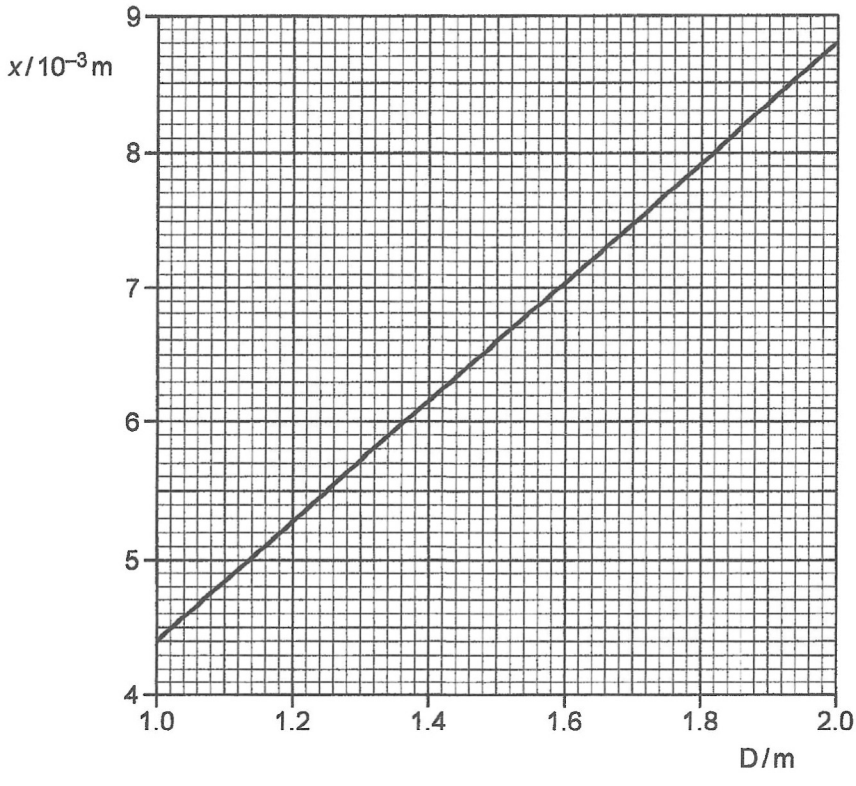


Fig. 4.1

The separation a of the slits is 0.12 mm.

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(i) Calculate the wavelength λ of the laser light.

λ = ... m [2]

(ii) A filter is used so that the amplitude of the light from one slit is halved.

Calculate the ratio

$$\frac{\text{maximum intensity of central bright fringe}}{\text{maximum intensity of adjacent dark fringe}}.$$

ratio = ... [3]

[Total: 8]

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5 (a) A silver wire of length 0.42 m has diameter 0.38 mm. The number density of free electrons in silver is $5.9 \times 10^{28}\,m^{-3}$. The drift velocity of electrons in the wire is $7.2 \times 10^{-5}\,m\,s^{-1}$.

Calculate the charge Q passing a point in the wire in a time of 30 minutes.

$Q =$ ..... C [2]

(b) An a.c. power supply is connected to three resistors, as shown in Fig. 5.1.

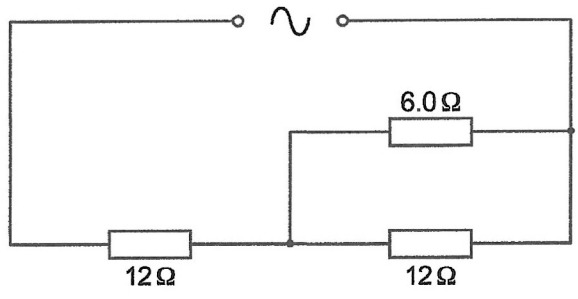


Fig. 5.1

The variation with time t of the voltage V of the power supply is given by the expression shown.

$$V = 9.0 \sin 120\pi t$$

(i) For the power supply determine:

1. the frequency f

$f =$ ..... Hz [1]

2. the root-mean-square (r.m.s.) voltage V_{rms}.

$V_{rms} =$ ..... V [1]

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(ii) 1. Calculate the peak current I_o from the power supply.

I_o = ... A [3]

2. Calculate the mean power P dissipated in the resistor of resistance 6.0 Ω.

P = ... W [2]

[Total: 9]

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6 **(a)** Define *magnetic flux density*.

.....

..... [1]

(b) A long solenoid of diameter 4.0 cm has 520 turns of insulated copper wire. The diameter of the wire is 0.46 mm. The wire is wrapped in a single layer with no gaps between each of the turns. The copper wire has resistivity $1.7 \times 10^{-8}\,\Omega\,m$.

Calculate the resistance R of the wire of the solenoid.

R = ... Ω [3]

(c) The solenoid in **(b)** is connected to a 24 V supply.

Use information from **(b)** to calculate the magnetic flux density B along the axis of the solenoid, at its centre.

B = ... T [3]

(d) A current-carrying wire is placed inside, and along the axis of the solenoid in **(b)**. The solenoid remains connected to the 24 V supply.

State and explain the magnitude of the force, if any, experienced by the wire.

.....

..... [1]

[Total: 8]

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7 **(a)** Explain what is meant by a *photon*.

.....

.....

..... [2]

(b) A beam of white light passes through a cloud of gas. The spectrum of the transmitted light contains a number of dark lines.

Explain why these dark lines occur.

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..... [4]

(c) Some electron energy levels in a hydrogen atom are illustrated in Fig. 7.1.

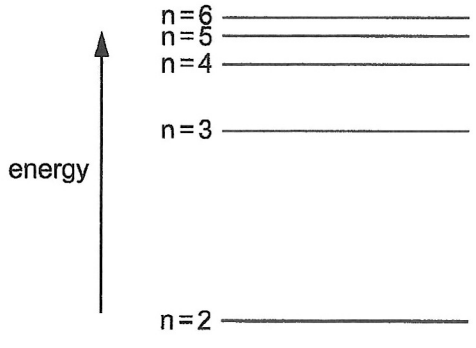


Fig. 7.1

The energy E in eV of each energy level may be determined using the expression

$$E = \frac{-13.6}{n^2}$$

where n is the energy level.

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(i) Calculate the energy, in eV, of energy level $n = 2$.

energy = ... eV [1]

(ii) Determine the wavelength λ of light absorbed when an electron in energy level $n = 2$ is excited to energy level $n = 3$.

λ = ... m [3]

(iii) Explain why the energy of each energy level is negative.

.....

..... [1]

[Total: 11]

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8 Read the passage below and answer the questions that follow.

Racing competitions, such as Formula 1, have led to advances in technology that eventually make their way into road vehicles. Formula 1 teams compete to make the fastest vehicle possible within strict rules.

Formula 1 drivers experience large accelerations and decelerations. A Formula 1 car can stop in a straight-line distance of 80 m from a maximum speed of 185 km h^{-1}.

High accelerations and decelerations are limited by the friction F between the tyres and the road. This is related to the weight W of the car by the equation

$$F = \mu W$$

where μ is the coefficient of friction.

Before use in a race, tyres are kept warm in special tyre blankets. The coefficient of friction varies with the tyre surface temperature, as shown in Fig. 8.1.

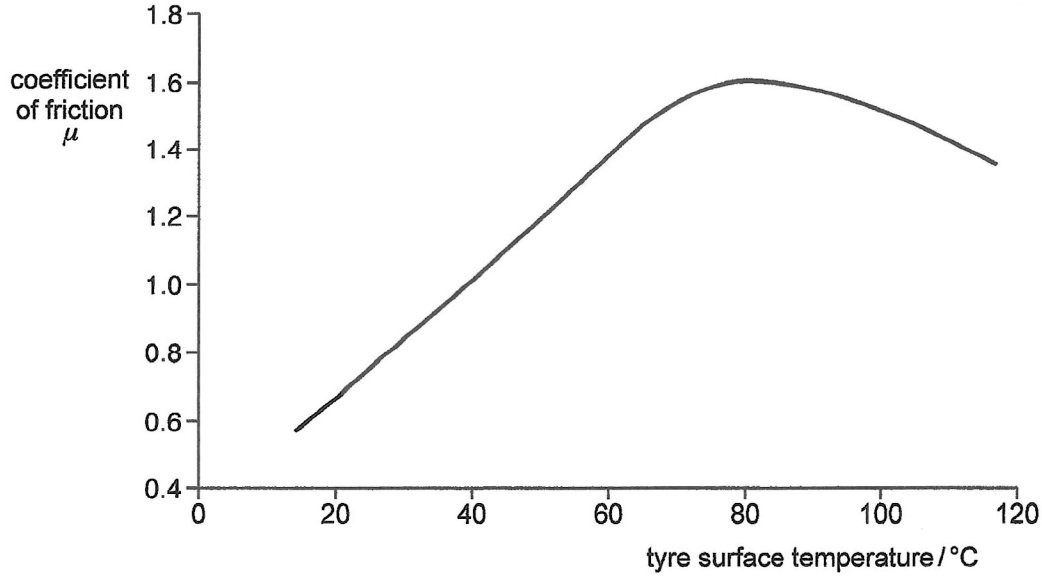


Fig. 8.1

A section through the road surface and a tyre at 20 °C in contact with the road is shown in Fig. 8.2.

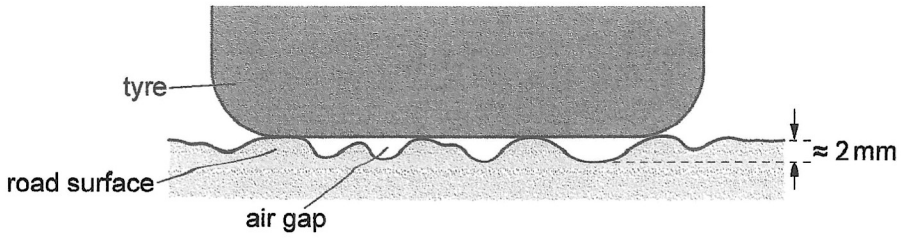


Fig. 8.2 (not to scale)

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When the tyre is heated with the tyre blanket, the compound of the tyre becomes softer, increasing the contact area.

In order to make the tightest circular corner in the Singapore Street Circuit, drivers decelerate from 220 km h^{-1} to make it around the corner of radius 30 m. The tyres have to withstand lateral (sideways) accelerations of up to 4*g* during cornering.

The mass of a typical Formula 1 car is about 750 kg. Greater accelerations and cornering speeds may be achieved with greater friction between the tyres and the track. Features of the car, such as rear wings, appear to increase the weight of the car.

When a Formula 1 car accelerates, the weight distribution between the front and rear wheels changes. The cars are rear-wheel drive, which means the accelerating force is provided through the rear wheels. When accelerating, the contact force experienced by the rear wheels increases and this is called 'weight transfer'.

The brakes used on Formula 1 cars consist of carbon fibre discs and pads. There is one set for each of the four wheels. Each set has a mass of 1.2 kg. There are at least 1000 small holes drilled into each disc. These brakes operate more effectively at high temperatures than materials used for the brakes on road vehicles. It is predicted that brakes made from carbon fibre discs and pads will be the next big change to reach the everyday car market.

(a) Calculate the maximum deceleration a Formula 1 driver experiences while braking to rest in a straight line.

deceleration = ... $m\,s^{-2}$ [2]

(b) (i) Determine the maximum speed a Formula 1 car can travel around the tightest circular corner at the Singapore Street Circuit.

maximum speed = ... $m\,s^{-1}$ [3]

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(ii) By reference to Fig. 8.1 and Fig. 8.2, suggest and explain why the tyres are heated before use.

.....

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.....

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..... [3]

(c) A cross-section of the rear wing on a Formula 1 car is shown in Fig. 8.3.

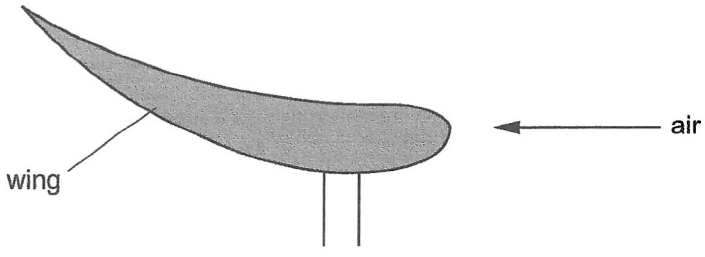


Fig. 8.3

The wing deflects air when the car is moving.

Explain, by reference to the change in momentum of the air, why this air flow appears to increase the weight of the car.

.....

.....

.....

.....

.....

..... [2]

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(d) Some of the forces acting on a Formula 1 car while accelerating from rest are shown in Fig. 8.4.

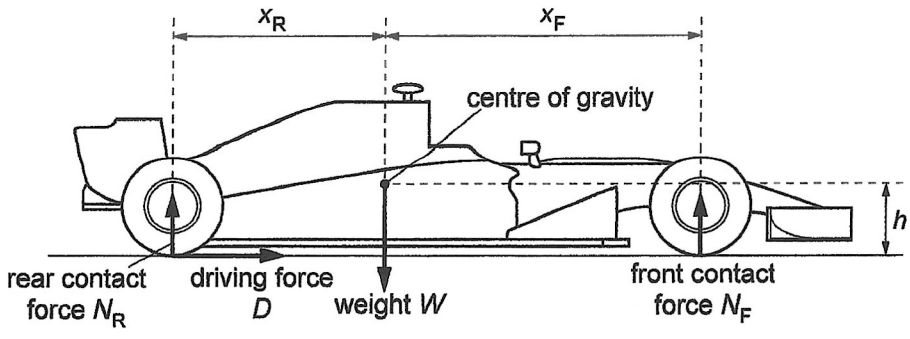


Fig. 8.4

N_R and N_F are the contact forces acting on the rear and front wheels, respectively. W is the weight of the car. D is the driving force acting on the rear wheels.

The centre of gravity is a vertical distance h above the ground. The centre of gravity is a horizontal distance x_R from the rear wheels and a horizontal distance x_F from the front wheels.

For the car accelerating from rest:

(i) take moments about the centre of gravity, to show that

$$N_R = \frac{Wx_F + Dh}{x_F + x_R}$$

[3]

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